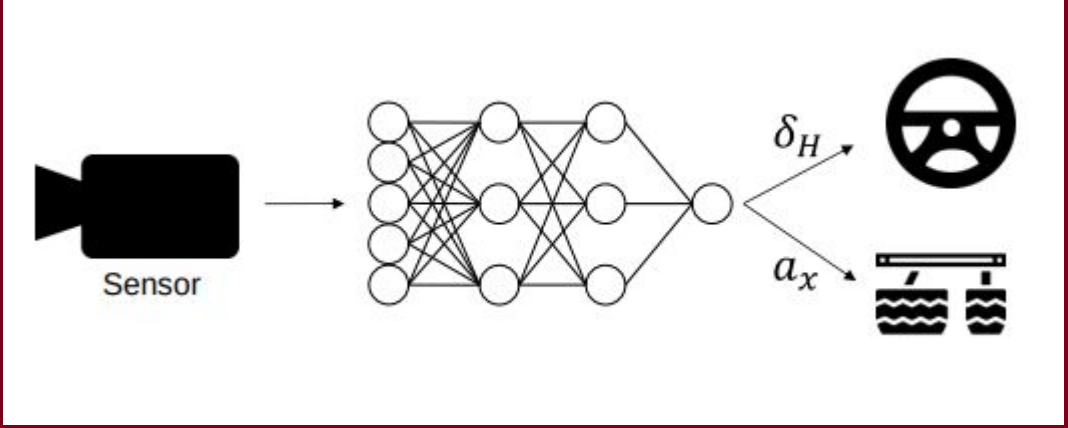
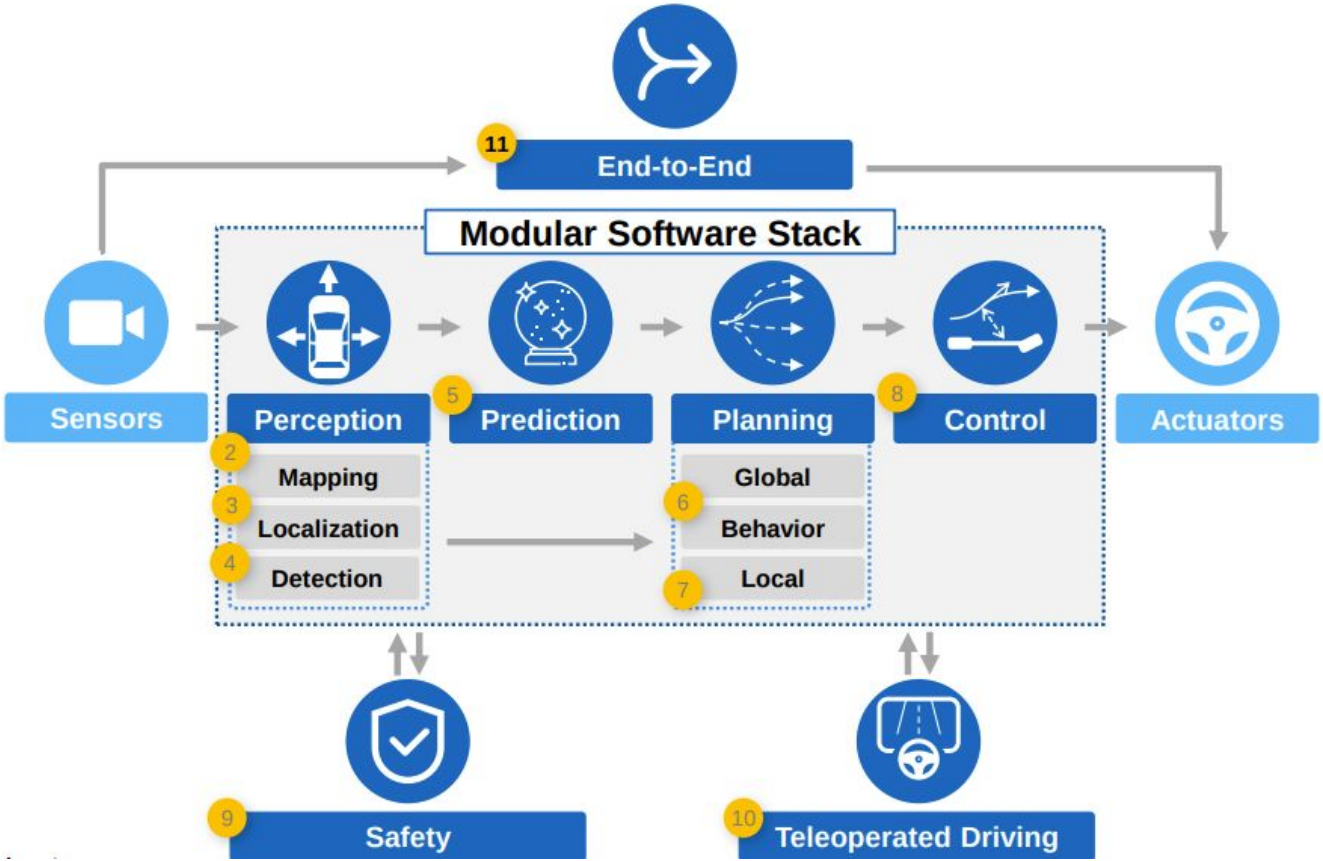
End to End driving using VLA

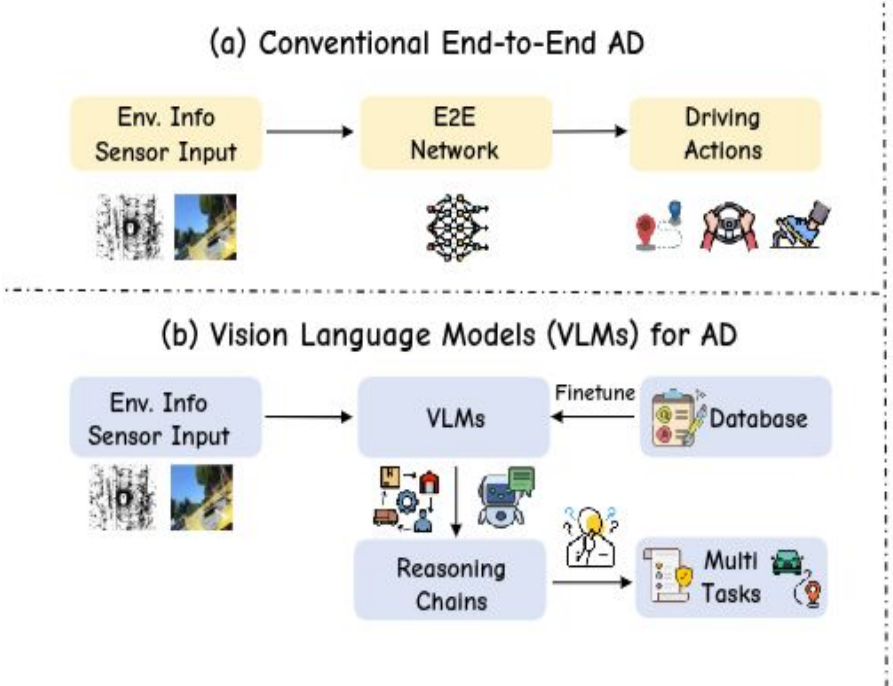


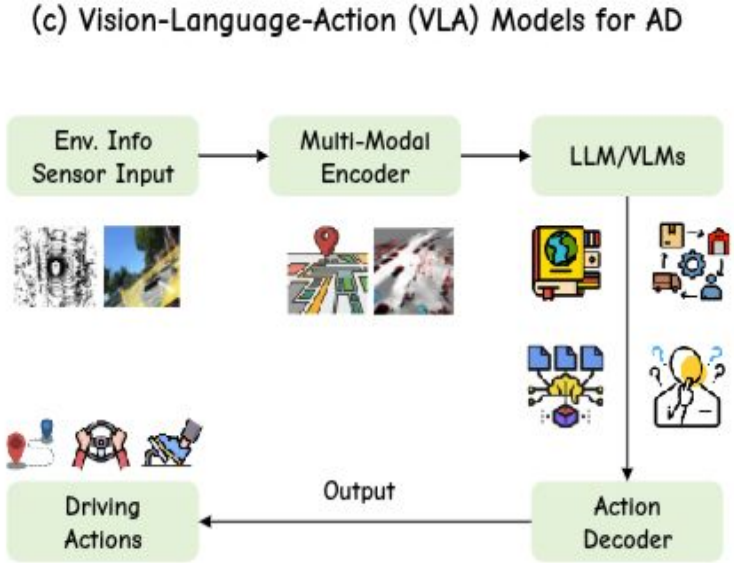
Introduction

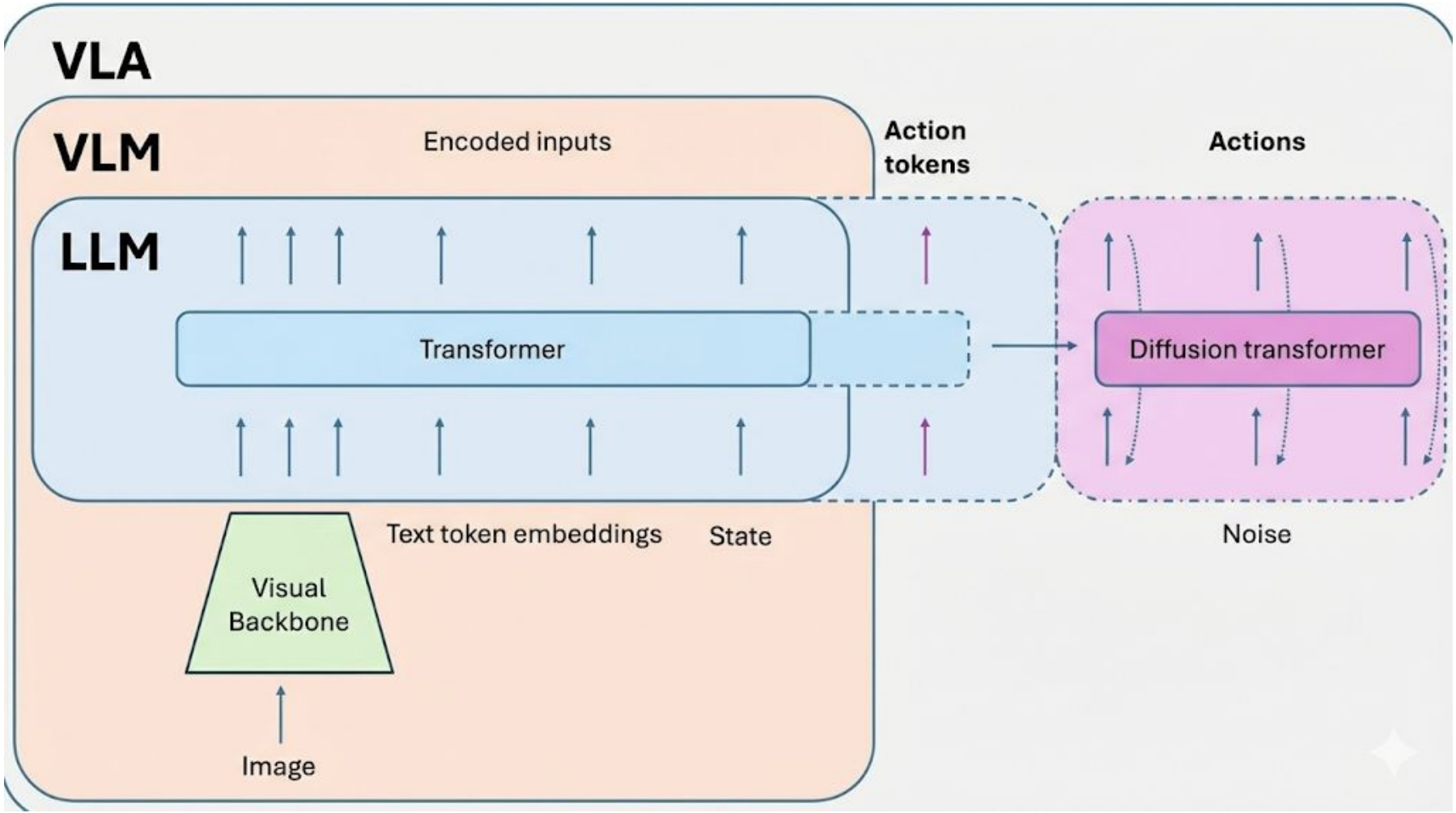


Ways to solve End to End driving for Autonomous Vehicle

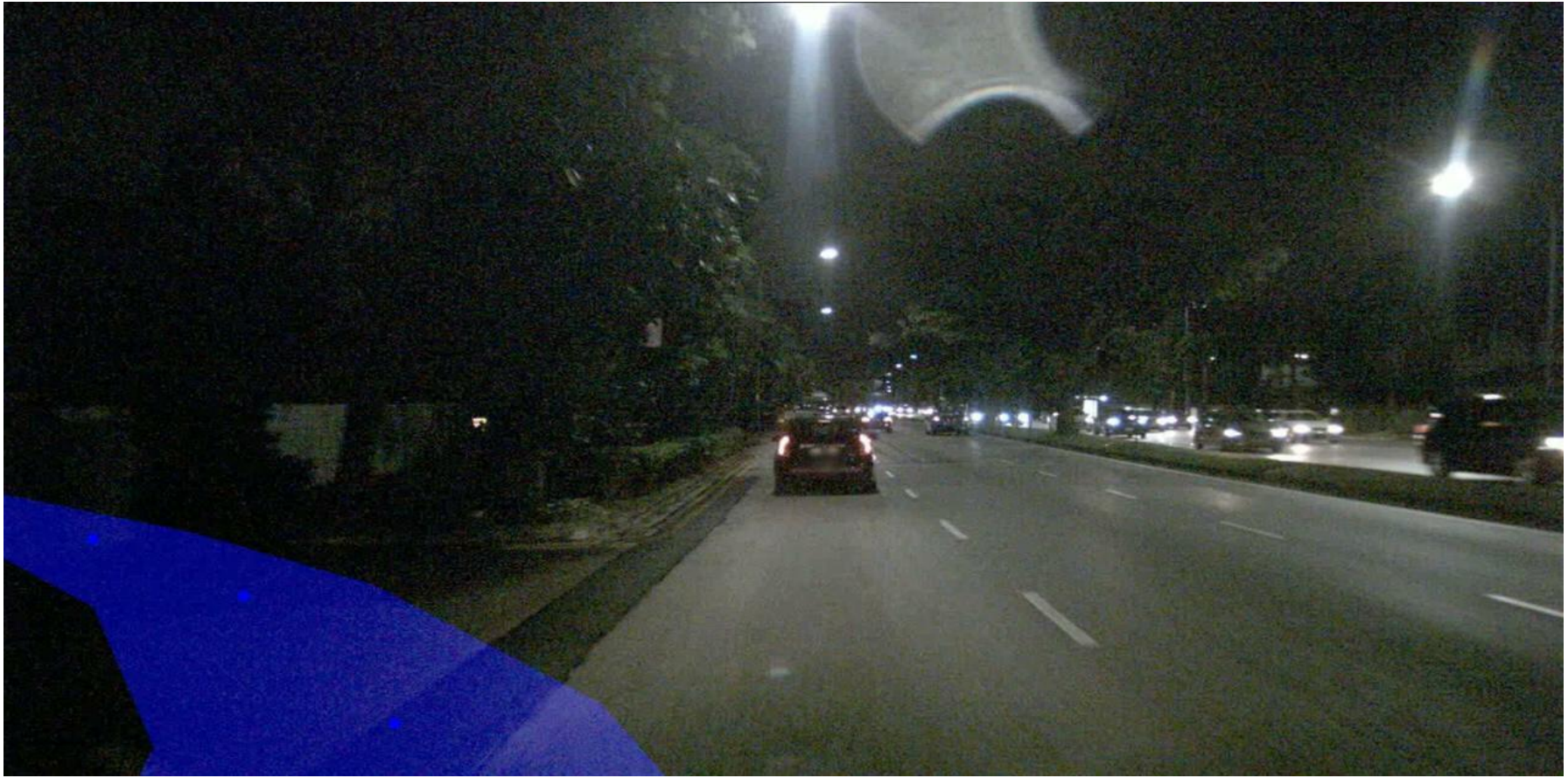
1- Imitation Learning(behaviour cloning)
2- VLM as the main planner
3- Vision Action models (most effective, we are going to use this)

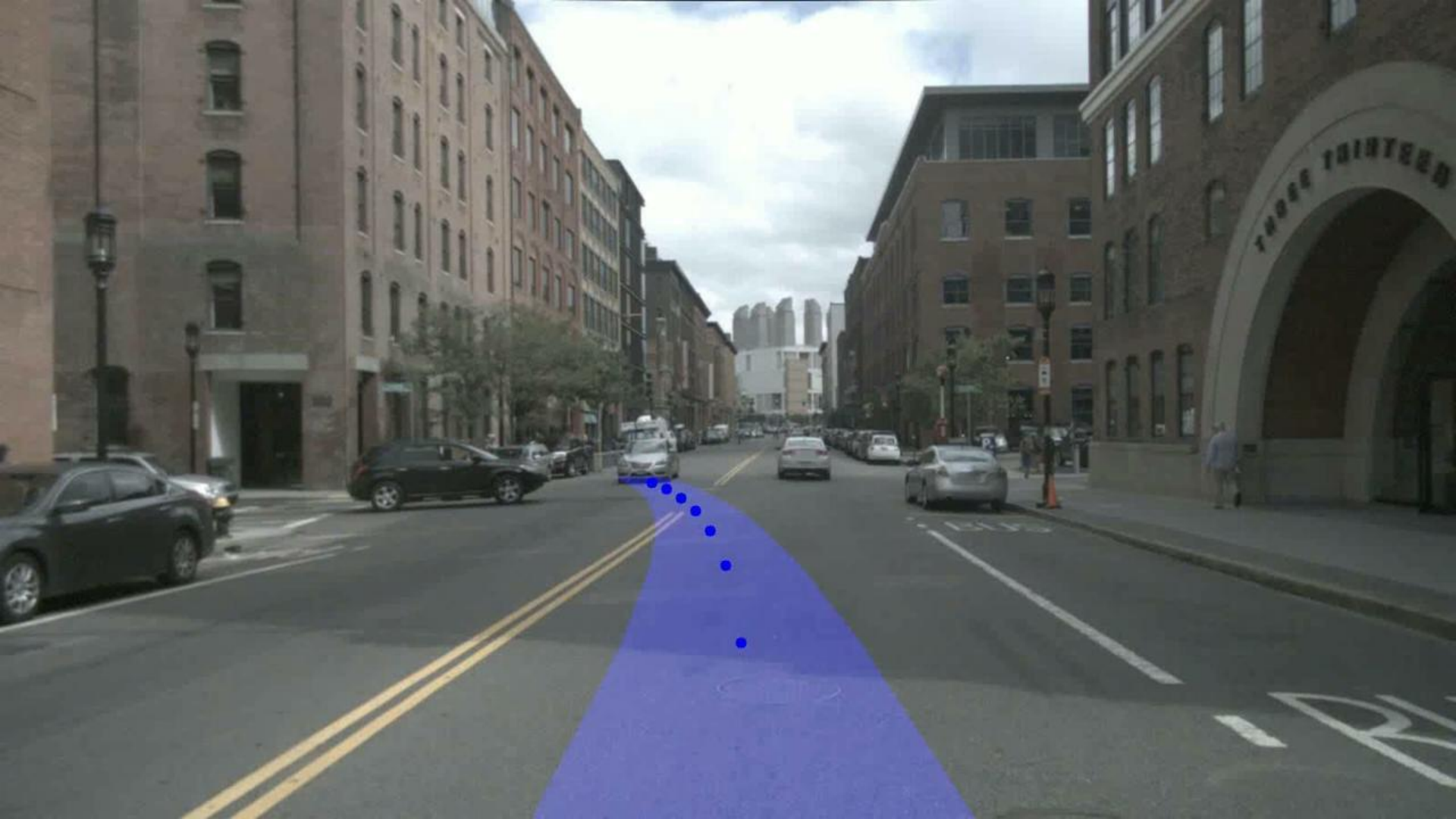





VLA
VLM
Encoded inputs
Action tokens
Actions
LLM
Transformer
Diffusion transformer
Text token embeddings
State
Noise
Visual Backbone
Image

Results on Nuscene mini Dataset (10 - 20 seconds scene only)

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Results on our Rover

Model used:-

Gemini 2.0 flash with fine tuning